

Atividade Pratica - API de Filmes

Node.js + Fastify

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Disciplina: Projeto e Desenvolvimento de Sistemas Web

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1. Estrutura das Pastas

Organizacao do projeto com rotas, plugins, schemas e servico de integracao externa.

```
✓ src
  ✓ plugins
    JS requestLogger.js
    JS swagger.js
  ✓ routes
    JS filmes.js
    JS home.js
  ✓ schemas
    JS movieSchemas.js
  ✓ services
    JS ghibliApi.js
```

2. Servidor Fastify em Execucao

Terminal mostrando o projeto iniciado e o servidor escutando em `http://127.0.0.1:3000`.

```
lucassilvamaues@MacBook-Air-de-Lucas ~ % /Users/lucassilvamaues/Documents/Web/Tarefa\ 9/abrir_tarefa_9.command ; exit;

up to date, audited 87 packages in 712ms

37 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

> tarefa-9-api-filmes-fastify@1.0.0 dev
> node --watch-path=src src/server.js

{"level":30,"time":1780903347596,"pid":55933,"hostname":"MacBook-Air-de-Lucas.local","msg":"Server listening at http://127.0.0.1:3000"}
{"level":30,"time":1780903347596,"pid":55933,"hostname":"MacBook-Air-de-Lucas.local","msg":"Server listening at http://192.168.1.3:3000"}
```

3. Interface de Validacao - Swagger UI

Swagger UI com as rotas disponiveis para teste no navegador.

The screenshot displays the Swagger UI for an API titled "Tarefa 9 - API de Filmes". The interface is dark-themed. At the top, the "fastify" logo is on the left, and the URL "http://127.0.0.1:3000/docs/json" is in the center, with an "Explore" button to its right. Below the title, the version "1.0.0" and "OAS 3.0" are shown. A description reads: "API REST com Node.js, Fastify, hooks, schemas e consumo de API externa." The main content is organized into three expandable sections: "status" (Verificacao da API), "filmes" (Rotas de filmes e busca), and "Schemas". The "status" section contains one endpoint: GET / (Verifica se a API esta funcionando). The "filmes" section contains three endpoints: GET /filmes (Lista filmes da API publica), GET /filmes/{id} (Busca detalhes de um filme pelo ID), and GET /busca (Pesquisa filmes pelo titulo). The "Schemas" section lists two schemas: filmeResumo and filmeDetalhe, each with a right-pointing arrow.

fastify  <http://127.0.0.1:3000/docs/json> **Explore** 

Tarefa 9 - API de Filmes

1.0.0 OAS 3.0

<http://127.0.0.1:3000/docs/json>

API REST com Node.js, Fastify, hooks, schemas e consumo de API externa.

status

Verificacao da API

GET / Verifica se a API esta funcionando

filmes

Rotas de filmes e busca

GET /filmes Lista filmes da API publica

GET /filmes/{id} Busca detalhes de um filme pelo ID

GET /busca Pesquisa filmes pelo titulo

Schemas

filmeResumo >

filmeDetalhe >

4. Rota GET /

Rota inicial retornando a mensagem de funcionamento da API.

GET / Verifica se a API esta funcionando

Parameters Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:3000/' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:3000/
```

Server response

Code	Details
200	Response body <pre>{ "mensagem": "API de filmes funcionando!" }</pre> Download Response headers <pre>access-control-allow-origin: * connection: keep-alive content-length: 41 content-type: application/json; charset=utf-8 date: Mon, 08 Jun 2026 08:34:30 GMT keep-alive: timeout=72</pre>

5. Rota GET /filmes

Rota consumindo a API publica e retornando uma lista de filmes.

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:3000/filmes' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:3000/filmes
```

Server response

Code	Details
200	Response body<pre>[{ "id": "86e544fd-79de-4e04-be62-5be67d8dd92e", "titulo": "Nausicaã of the Valley of the Wind", "tituloOriginal": "風の谷のナウシカ", "diretor": "Hayao Miyazaki", "anoLancamento": "1984", "nota": "89" }, { "id": "2baf70d1-42bb-4437-b551-e5fed5a87abe", "titulo": "Castle in the Sky", "tituloOriginal": "天空の城ラピュタ", "diretor": "Hayao Miyazaki", "anoLancamento": "1986", "nota": "96" }, { "id": "12cfb892-aac0-4c5b-94af-521852e46d6a", "titulo": "Grave of the Fireflies", "tituloOriginal": "火垂るの墓", "diretor": "Isao Takahata", "anoLancamento": "1988", "nota": "100" }]</pre> Response headers<pre>access-control-allow-origin: * connection: keep-alive content-length: 4516 content-type: application/json; charset=utf-8 date: Mon,08 Jun 2026 08:35:13 GMT keep-alive: timeout=72</pre>

6. Rota GET /filmes/:id

Rota com parametro de URL retornando os detalhes de um filme.

GET `/filmes/{id}` Busca detalhes de um filme pelo ID

Parameters

Name

Description

id * required
string
(path)

58611129-2dbc-4a81-a72f-77ddfc1b1b49

Execute

Clear

Responses

Curl

```
curl -X 'GET' \  
'http://127.0.0.1:3000/filmes/58611129-2dbc-4a81-a72f-77ddfc1b1b49' \  
-H 'accept: application/json'
```

Request URL

http://127.0.0.1:3000/filmes/58611129-2dbc-4a81-a72f-77ddfc1b1b49

Server response

Code	Details
200	Response body <pre>{ "id": "58611129-2dbc-4a81-a72f-77ddfc1b1b49", "titulo": "My Neighbor Totoro", "tituloOriginal": "となりのトトロ", "descricao": "Two sisters move to the country with their father in order to be closer to their hospitalized mother, and discover the surrounding trees are inhabited by Totoros, magical spirits of the forest. When the youngest runs away from home, the older sister seeks help from the spirits to find her.", "diretor": "Hayao Miyazaki", "produtor": "Hayao Miyazaki", "anoLancamento": "1988", "duracao": "86", "nota": "93" }</pre>

6. Rota GET /filmes/:id - continuacao

Continuacao do retorno da rota com parametro de URL.

Response headers

```
access-control-allow-origin: *
connection: keep-alive
content-length: 526
content-type: application/json; charset=utf-8
date: Mon, 08 Jun 2026 07:28:29 GMT
keep-alive: timeout=72
```

Responses

Code	Description	Links
200	Default Response Media type application/json Controls Accept header. Example Value Schema <pre>{ "id": "string", "titulo": "string", "tituloOriginal": "string", "descricao": "string", "diretor": "string", "produtor": "string", "anoLancamento": "string", "duracao": "string", "nota": "string" }</pre>	

7. Rota GET /busca?q=totoro

Rota com query parameter pesquisando filmes pelo titulo.

GET **/busca** Pesquisa filmes pelo titulo

Parameters

Cancel

Name	Description
q * required string (query)	totoro

Execute

Clear

Responses

Curl

```
curl -X 'GET' \  
  'http://127.0.0.1:3000/busca?q=totoro' \  
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:3000/busca?q=totoro
```

Server response

Code	Details
200	Response body <pre>[{ "id": "58611129-2dbc-4a81-a72f-77ddfc1b1b49", "titulo": "My Neighbor Totoro", "tituloOriginal": "となりのトトロ", "diretor": "Hayao Miyazaki", "anoLancamento": "1988", "nota": "93" }]</pre> Download

7. Rota GET /busca?q=totoro - continuacao

Continuacao do retorno da busca pelo termo totoro.

Response headers

```
access-control-allow-origin: *
connection: keep-alive
content-length: 180
content-type: application/json; charset=utf-8
date: Mon,08 Jun 2026 07:28:50 GMT
keep-alive: timeout=72
```

Responses

Code	Description	Links
200	Default Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
[
  {
    "id": "string",
    "titulo": "string",
    "tituloOriginal": "string",
    "diretor": "string",
    "anoLancamento": "string",
    "nota": "string"
  }
]
```

8. Middleware Registrando Logs

Hook exibindo no terminal o metodo HTTP, URL acessada e horario da requisicao.

```
{
  "level": 30,
  "time": 1780903615220,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-1",
  "req": {
    "method": "GET",
    "url": "/docs",
    "host": "127.0.0.1:3000",
    "remoteAddress": "127.0.0.1",
    "remotePort": 57467
  },
  "msg": "incoming request"
}
{
  "level": 30,
  "time": 1780903615243,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "metodo": "GET",
  "url": "/docs",
  "horario": "08/06/2026, 04:26:55",
  "msg": "Requisicao recebida"
}
{
  "level": 30,
  "time": 1780903615247,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-1",
  "res": {
    "statusCode": 200,
    "responseTime": 26.66404104232788,
    "msg": "request completed"
  }
}
{
  "level": 30,
  "time": 1780903615587,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-2",
  "req": {
    "method": "GET",
    "url": "/docs/static/swagger-initializer.js",
    "host": "127.0.0.1:3000",
    "remoteAddress": "127.0.0.1",
    "remotePort": 57467,
    "msg": "incoming request"
  }
}
{
  "level": 30,
  "time": 1780903615588,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "metodo": "GET",
  "url": "/docs/static/swagger-initializer.js",
  "horario": "08/06/2026, 04:26:55",
  "msg": "Requisicao recebida"
}
{
  "level": 30,
  "time": 1780903615588,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-2",
  "res": {
    "statusCode": 200,
    "responseTime": 1.4435830116271973,
    "msg": "request completed"
  }
}
{
  "level": 30,
  "time": 1780903615591,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-3",
  "req": {
    "method": "GET",
    "url": "/docs/static/index.css",
    "host": "127.0.0.1:3000",
    "remoteAddress": "127.0.0.1",
    "remotePort": 57468,
    "msg": "incoming request"
  }
}
{
  "level": 30,
  "time": 1780903615592,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "metodo": "GET",
  "url": "/docs/static/index.css",
  "horario": "08/06/2026, 04:26:55",
  "msg": "Requisicao recebida"
}
{
  "level": 30,
  "time": 1780903615594,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-4",
  "req": {
    "method": "GET",
    "url": "/docs/static/swagger-ui.css",
    "host": "127.0.0.1:3000",
    "remoteAddress": "127.0.0.1",
    "remotePort": 57469,
    "msg": "incoming request"
  }
}
{
  "level": 30,
  "time": 1780903615595,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "metodo": "GET",
  "url": "/docs/static/swagger-ui.css",
  "horario": "08/06/2026, 04:26:55",
  "msg": "Requisicao recebida"
}
{
  "level": 30,
  "time": 1780903615595,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-5",
  "req": {
    "method": "GET",
    "url": "/docs/static/swagger-ui-standalone-preset.js",
    "host": "127.0.0.1:3000",
    "remoteAddress": "127.0.0.1",
    "remotePort": 57470,
    "msg": "incoming request"
  }
}
{
  "level": 30,
  "time": 1780903615596,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "metodo": "GET",
  "url": "/docs/static/swagger-ui-standalone-preset.js",
  "horario": "08/06/2026, 04:26:55",
  "msg": "Requisicao recebida"
}
{
  "level": 30,
  "time": 1780903615596,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-6",
  "req": {
    "method": "GET",
    "url": "/docs/static/swagger-ui-bundle.js",
    "host": "127.0.0.1:3000",
    "remoteAddress": "127.0.0.1",
    "remotePort": 57471,
    "msg": "incoming request"
  }
}
{
  "level": 30,
  "time": 1780903615596,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "metodo": "GET",
  "url": "/docs/static/swagger-ui-bundle.js",
  "horario": "08/06/2026, 04:26:55",
  "msg": "Requisicao recebida"
}
{
  "level": 30,
  "time": 1780903615600,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-3",
  "res": {
    "statusCode": 304,
    "responseTime": 8.790624976158142,
    "msg": "request completed"
  }
}
{
  "level": 30,
  "time": 1780903615602,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-4",
  "res": {
    "statusCode": 304,
    "responseTime": 7.064333021640778,
    "msg": "request completed"
  }
}
{
  "level": 30,
  "time": 1780903615602,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-5",
  "res": {
    "statusCode": 304,
    "responseTime": 6.570958971977234,
    "msg": "request completed"
  }
}
{
  "level": 30,
  "time": 1780903615602,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-6",
  "res": {
    "statusCode": 304,
    "responseTime": 6.207709014415741,
    "msg": "request completed"
  }
}
{
  "level": 30,
  "time": 1780903615719,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-7",
  "req": {
    "method": "GET",
    "url": "/docs/json",
    "host": "127.0.0.1:3000",
    "remoteAddress": "127.0.0.1",
    "remotePort": 57467,
    "msg": "incoming request"
  }
}
{
  "level": 30,
  "time": 1780903615719,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "metodo": "GET",
  "url": "/docs/json",
  "horario": "08/06/2026, 04:26:55",
  "msg": "Requisicao recebida"
}
{
  "level": 30,
  "time": 1780903615725,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-7",
  "res": {
    "statusCode": 200,
    "responseTime": 5.770416021347046,
    "msg": "request completed"
  }
}
{
  "level": 30,
  "time": 1780903615725,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-8",
  "req": {
    "method": "GET",
    "url": "/docs/static/favicon-16x16.png",
    "host": "127.0.0.1:3000",
    "remoteAddress": "127.0.0.1",
    "remotePort": 57468,
    "msg": "incoming request"
  }
}
{
  "level": 30,
  "time": 1780903615726,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "metodo": "GET",
  "url": "/docs/static/favicon-16x16.png",
  "horario": "08/06/2026, 04:26:55",
  "msg": "Requisicao recebida"
}
{
  "level": 30,
  "time": 1780903615727,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-8",
  "res": {
    "statusCode": 304,
    "responseTime": 1.396416962146759,
    "msg": "request completed"
  }
}
{
  "level": 30,
  "time": 1780903709140,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-9",
  "req": {
    "method": "GET",
    "url": "/filmes/58611129-2dbc-4a81-a72f-77ddfc1b1b49",
    "host": "127.0.0.1:3000",
    "remoteAddress": "127.0.0.1",
    "remotePort": 57484,
    "msg": "incoming request"
  }
}
{
  "level": 30,
  "time": 1780903709141,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "metodo": "GET",
  "url": "/filmes/58611129-2dbc-4a81-a72f-77ddfc1b1b49",
  "horario": "08/06/2026, 04:28:29",
  "msg": "Requisicao recebida"
}
{
  "level": 30,
  "time": 1780903709303,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-9",
  "res": {
    "statusCode": 200,
    "responseTime": 163.1704579591751,
    "msg": "request completed"
  }
}
{
  "level": 30,
  "time": 1780903730370,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-a",
  "req": {
    "method": "GET",
    "url": "/busca?q=totoro",
    "host": "127.0.0.1:3000",
    "remoteAddress": "127.0.0.1",
    "remotePort": 57484,
    "msg": "incoming request"
  }
}
{
  "level": 30,
  "time": 1780903730370,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "metodo": "GET",
  "url": "/busca?q=totoro",
  "horario": "08/06/2026, 04:28:50",
  "msg": "Requisicao recebida"
}
{
  "level": 30,
  "time": 1780903730471,
  "pid": 55933,
  "hostname": "MacBook-Air-de-Lucas.local",
  "reqId": "req-a",
  "res": {
    "statusCode": 200,
    "responseTime": 100.9550409913063,
    "msg": "request completed"
  }
}
```

9. Conclusao

A API desenvolvida atende aos requisitos da atividade: servidor Fastify, rotas HTTP, parametro de rota, query parameter, organizacao das rotas, middleware com hook, consumo de API externa e tratamento de erros.

Observacao: o Swagger UI foi usado como interface de validacao das rotas, evitando a criacao manual de uma tela apenas para testes.